

Gunimi — AI-First Workspace Operating System

Version 1.0 — Confidential

Executive Summary

Gunimi is an AI-native workspace operating system built for modern businesses. It replaces the fragmented stack of CRM tools, project management apps, email clients, and note-taking systems with a single, intelligent workspace that understands the full context of a business — its people, relationships, deals, tasks, communication, and signals.

Gunimi is not a CRM. It is not a Notion clone. It is a living workspace that learns, observes, and surfaces intelligence automatically — so teams can focus on relationships and execution, not on data entry.

The Problem

Modern business teams operate across 5–10 disconnected tools simultaneously:

- **CRM** (HubSpot, Pipedrive) — contacts and deals, but no real intelligence
- **Project management** (Linear, Asana, Monday) — tasks and execution, isolated from relationships
- **Note-taking** (Notion, Obsidian) — knowledge, disconnected from context
- **Email** (Gmail, Outlook) — communication, invisible to the rest of the stack
- **Analytics** (dashboards) — reactive, not proactive

The result: context is lost at every handoff. Deals stall silently. Relationships decay unnoticed. Teams spend more time managing tools than managing work.

The Solution

Gunimi unifies the entire business context into one workspace:

Companies → Contacts → Deals → Tasks → Notes → Email → Intelligence

Every entity connects to every other. Every interaction generates a signal. Every signal feeds the intelligence layer. The workspace learns over time and surfaces what matters — without being asked.

Core Principles

- **Context first** — every piece of information is linked to people, companies, and deals
 - **Intelligence by default** — the system observes and surfaces, the user doesn't configure
 - **AI native** — not AI added on top, but AI woven into every layer
 - **Minimal surface** — premium, enterprise-grade UI with no visual clutter
 - **Workspace, not app** — a living environment, not a static tool
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Core Modules

1. Contacts (Relationship Management)

Full contact lifecycle management built on `workspace_people` — Gunimi's core people graph.

Features:

- Complete contact profiles (name, email, phone, position, company, status, notes)
- Priority flagging with visual indicators
- Status lifecycle: Lead → Active → Won
- Tag system with color-coded labels and AI-generated tag intelligence
- Bulk operations: multi-select, bulk tag, bulk delete, merge duplicates
- CSV import
- **Relationship Health Score** — automatic 0–100 score per contact based on:
 - Email recency and frequency (last 30 days)
 - Active deals
 - Open tasks
 - Tiers: Healthy (≥ 60) / Watching (30–59) / At Risk (10–29) / Cold (0–9)

2. Companies

Company management with full relationship graph:

- Company profiles (name, industry, website, size, description)
- Linked contacts, deals, tasks, notes, orders
- Signal integration — company-level activity tracking
- Tag support and AI intelligence

3. Deals (Pipeline Management)

Sales pipeline with structured deal lifecycle:

- Deal title, value (EUR), stage, priority, linked contact and company
- Kanban + list views
- Stage tracking: Prospect → Qualified → Proposal → Negotiation → Closed Won / Lost
- Signal generation on deal creation, stage change, stall detection
- Full order history per deal

4. Tasks (Execution Layer)

Workspace-wide task management:

- Priority levels: High / Medium / Low
- Status: To Do / In Progress / Done
- Due dates with overdue detection
- Linked to contacts, companies, deals
- Assignee support across team members
- Signals generated for overdue tasks and completion events
- ⚡ Quick create from anywhere in the workspace

5. Notes (Knowledge Layer)

Structured knowledge capture with rich text support:

- Tiptap-based rich text editor
- #tag and @member mention system with hover cards
- Linked to contacts, companies, deals
- Workspace-wide notes feed
- ⌘K quick create

6. Email (Communication Layer)

Gmail-native email integration — read and write:

- OAuth 2.0 connection (Gmail API — approved)
- Automatic thread sync and CRM linking (threads matched to contacts/companies by email address)
- **Email Command Center** — unified inbox with:
 - Unread priority widget
 - Follow-up needed detection (heuristic: read thread, linked contact, no reply in 3+ days)
 - Recent threads
 - Linked contacts and companies ranked by thread volume
 - Email intelligence widget (CRM coverage rate, unread rate, top contacts/companies)
- **Email write (V2):**
 - Reply to thread from within Gunimi
 - Compose new email to any contact
 - AI-assisted draft based on relationship context (contact history, open deals, tasks)
 - Automatic Signal generation on email send
 - Reconnect flow for scope upgrades (gmail.send)

7. Orders

Business order management linked to contacts, deals, companies:

- Order title, value, currency (EUR default), status
- State machine: Draft → Confirmed → Shipped → Delivered / Cancelled / Refunded
- Signal generation on order creation, confirmation, fulfillment
- ⌘K quick create, Email → Order conversion
- Full order history per contact, deal, and company

8. Tags

Cross-entity tagging system with AI intelligence:

- Color-coded labels (8 color variants)
- Applied to contacts, companies, deals, tasks, notes
- Tag detail page: all tagged entities in one view
- **AI Tag Intelligence** — GPT-4o-mini analyzes all tagged entities and generates a 2–3 sentence business insight about the tag cluster
- TagHoverCard for inline preview
- Mention system: #tagname in comments and notes

Intelligence Layer

Signal Engine

The Signal Engine is Gunimi's core intelligence infrastructure — a real-time event processing layer that observes all workspace activity and generates actionable signals.

Architecture:

- 25+ signal types across all entity domains
- Signal lifecycle: **active** → **resolved** → **expired**
- Deduplication and suppression logic
- Signal identity: **signalId**, **workspaceId**, **origin**, **correlationId**, **parentSignalId**
- Signal evolution (a signal can change state while maintaining identity)
- Explainability: every signal answers 7 questions (what, who, when, why, what changed, what to do, what if ignored)

Signal categories:

- Relationship signals: contact went cold, follow-up needed, deal stalled
- Execution signals: task overdue, deal without tasks, no recent activity
- Communication signals: email sent, email received, no reply in N days
- Business signals: order confirmed, order shipped, deal won/lost
- Workspace signals: new member, new contact imported

Scan Engine:

- Vercel Cron: runs every 6 hours (**0 */6 * * ***)
- Scans all active workspaces
- Persists **lastRunAt** in workspace preferences
- **/api/signals/health** endpoint for monitoring
- Full observability: all scanners report real counts

Relationship Health Score

Automatic 0–100 health score per contact. Computed from 4 signal sources:

Signal	Max Points	Logic
Email recency	40	≤7d=40, ≤14d=32, ≤30d=22, ≤60d=12, ≤90d=5
Email frequency (30d)	25	5+=25, 3+=18, 1+=10
Total deals	20	2+=20, 1=14
Open tasks	15	2+=15, 1=10

Rendered as a colored dot + score in the CRM contacts list. Batch computed — 3 parallel DB queries for all contacts, aggregated in TypeScript.

AI Brief

Daily AI-generated workspace briefing:

- Summaries of open deals, overdue tasks, relationship signals
- Natural language, invisible AI (no "AI says" labels)
- Context-aware, workspace-specific

AI Tag Intelligence

Per-tag AI analysis using GPT-4o-mini:

- Analyzes all entities under a tag
- Detects patterns: industry clusters, deal stage concentration, stalled pipelines
- Cached for 24 hours, refreshable on demand
- Available in workspace language (EN/SK/CS)

Business Memory Blueprint

A documented architecture for persistent workspace intelligence:

- 7 memory types (contact, company, deal, interaction, preference, signal, workspace)
- Memory identity contract (full traceability: source, confidence, created_by, version)
- Memory versioning (immutable history)
- Memory graph: Memory ↔ Signal ↔ Story ↔ Workspace
- AI Trust Boundary: defined list of what AI may never store

Technical Architecture

Stack

Layer	Technology
Frontend	Next.js 15 (App Router, Server Components)
Language	TypeScript (strict)
Styling	TailwindCSS v4
Backend	Supabase (PostgreSQL + Row Level Security)
Auth	Supabase Auth (email/password, magic link)
AI	OpenAI GPT-4o-mini (via openai SDK)
Email	Gmail API (OAuth 2.0), Postmark (transactional)
Error tracking	Sentry
Deployment	Vercel (Edge + Serverless)
Internationalization	next-intl (EN / SK / CS)
Cron	Vercel Cron Jobs

Data Layer

```

workspace
├── workspace_people      (contacts – core people graph)
├── workspace_companies  (companies)
├── workspace_deals      (sales pipeline)
├── workspace_tasks      (execution layer)
├── workspace_notes      (knowledge)
├── workspace_orders     (business orders)
├── workspace_tags       (cross-entity labels)
├── workspace_teams      (organizational structure)
├── workspace_members    (user ↔ workspace membership)
├── email_connections    (Gmail OAuth tokens)
├── email_threads        (synced email threads)
├── email_messages       (individual messages)
├── signals              (intelligence events)
└── comments             (tiptap-based with mentions)

```

workspace_people is the primary people graph. **workspace_contacts** is a SQL VIEW for backwards compatibility.

Server Architecture

- **Server Actions** — all data mutations use Next.js Server Actions (no REST API for internal use)
- **supabaseAdmin** — server-side client with service role for all server actions
- **supabase (client)** — browser client with RLS-enforced user session for realtime
- **logger.ts** — server-side structured logging; zero **console.log** in production
- **Sentry** — all unhandled errors captured automatically; no **console.error** in client components

AI Platform Architecture

11-layer AI platform documented in [docs/blueprints/AI_PLATFORM_ARCHITECTURE.md](#):

1. Data Ingestion Layer
2. Signal Production Layer
3. Memory Layer
4. Context Assembly Layer
5. AI Budget Layer
6. Prompt Engineering Layer
7. Model Execution Layer
8. Response Processing Layer
9. Explainability Layer
10. Observability Layer
11. Trust & Safety Layer

AI Budget Controls:

- Per-workspace daily token limit (default: 100,000 tokens)
- Suspension kill switch (admin-level)
- **checkAIBudget()** called before every AI request
- Admin dashboard: risk bars, inline limit editor, suspend toggle

- Usage tracked per workspace per day
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Security

Authentication

- Supabase Auth with email/password and magic link
- Session management via httpOnly cookies
- Server-side session validation on every request

Row Level Security (RLS)

- All workspace data protected by PostgreSQL RLS policies
- Users can only access data within their workspace membership
- Service role (`supabaseAdmin`) used only in server actions — never exposed to client
- All client-side Supabase calls go through the session-bound client

Email OAuth

- Gmail OAuth tokens stored encrypted in `email_connections`
- Access tokens refreshed automatically before expiry (60s window)
- Token revocation detected and handled (credentials cleared)
- Scope upgrade flow with user-facing reconnect banner

Admin Console

- 9-section admin console with separate access control
- Audit log: every admin mutation logged with actor, action, timestamp, payload
- Broadcast system with banner + composer
- Workspace Manager with feature flags and AI budget controls
- User Manager with invite control

Production Standards

- Zero `console.log` / `console.debug` / `console.info` in production
- Zero `console.error` in client components
- All errors routed to Sentry
- Sentry DSN configured and active
- All HTTP-only cookies, no localStorage for sensitive data
- CSP headers configured

Content Security

- Dogfooding feedback system for internal quality monitoring
 - Runtime audit: no PGRST116 errors (all queries use `maybeSingle()` where 0 rows is valid)
 - No hydration errors, no failed suspense boundaries
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Scalability

Database

- PostgreSQL via Supabase — battle-tested, ACID compliant
- Connection pooling via Supabase connection pooler (PgBouncer)
- Indexes on all foreign keys and frequently queried columns
- RLS policies are performant (indexed on `workspace_id` + `user_id`)
- Horizontal scaling via read replicas (Supabase Pro+)

Application Layer

- Vercel Edge Network — global CDN for static assets
- Serverless functions — auto-scaling, no cold-start penalty for API routes
- Server Components — minimal client JS bundle, server-rendered by default
- `Promise.all()` pattern — parallel data fetching on every page (no waterfall)

AI Layer

- Per-workspace token budgets prevent runaway AI costs
- GPT-4o-mini — cost-efficient for high-frequency calls (tag intelligence, drafts)
- 24-hour cache on AI Tag Intelligence
- AI Brief generated on-demand, not on schedule

Email Sync

- Batch processing: 10 threads per batch (configurable)
- Incremental sync: `afterDate` filter on subsequent syncs
- Rate limit on sync endpoint (`ratelimit.ts`)
- Token refresh handled automatically in sync pipeline

Cron Jobs

- Signal scan: every 6 hours, all workspaces
- Workspace-level `lastRunAt` persisted — idempotent scans
- `CRON_SECRET` protected endpoint

Team Isolation (Planned)

- Current: all workspace members share full visibility (correct for SMB)
 - V2 plan: per-workspace `team_isolation` toggle
 - `false` (default): full shared visibility
 - `true`: RLS team-scoped filtering, activated per customer request
 - Designed so small collaborative teams are never affected by enterprise isolation features
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Internationalization

- **3 languages**: English (EN), Slovak (SK), Czech (CS)
- **2,800+ locale keys** covering every visible string in the application
- `next-intl` — server and client components both covered
- Server components: `const t = await getTranslations('namespace')`

- Client components: `const t = useTranslations('namespace')`
- Cookie-based locale switching with SSR support
- AI responses generated in workspace language (EN/SK/CS)
- Zero hardcoded user-facing strings in production

Admin Console

9-section platform administration dashboard at `/dashboard/admin`:

Section	Capability
Hub	Platform overview, key metrics
Workspace Manager	All workspaces, feature flags, AI budget
User Manager	All users, invite control
AI Ops	Per-workspace token limits, suspension, risk visualization
Platform Health	System status, cron health, error rates
Invite Control	Global invite settings, pending invites
Broadcast	Platform-wide banner + message composer
Audit Log	Full admin action history
Dogfood	Internal feedback dashboard

All admin mutations are logged to the audit system with actor, action, timestamp, and payload.

Design System (GDL — Gunimi Design Language)

Consistent, premium component library:

- `GunimiSection` — page section wrapper
- `GunimiHeading` — badge + title + subtitle pattern
- `GunimiCard` — card container with consistent border/bg
- `GunimiMetricCard` / `GunimiStatCard` — KPI display
- `GunimiButton` — primary / secondary / danger variants with loading state
- `GunimiInput` / `GunimiField` — form primitives
- `GunimiEmptyState` — consistent empty state pattern
- `GunimiWorkspaceTabs` — entity workspace navigation

Design tokens:

- Dark-first: `#080C14` base surface
- Violet primary: `#6D5BFF` / `#8B7DFF`
- Semantic colors: emerald (success), amber (warning), red (danger), zinc (neutral)
- Typography: Inter (system) — 10px tracking labels to 24px headings
- Border: `border-white/[0.07]` — consistent subtle separation

Product Roadmap

Completed (Beta)

-  Core CRM: Contacts, Companies, Deals
-  Tasks with assignees, priorities, due dates
-  Notes with rich text and mentions
-  Tags with AI Intelligence
-  Email Command Center (Gmail read)
-  Email V2 (Gmail write, reply, compose, AI draft)
-  Orders with signal integration
-  Signal Engine (25 signal types, scan engine)
-  Relationship Health Score
-  AI Brief
-  AI Budget Controls
-  Admin Console (9 sections)
-  Teams (organizational labels)
-  Domain Foundation (workspace_people graph)
-  Workspace Preferences (language, regional)
-  Internationalization (EN/SK/CS, 2800+ keys)
-  Sentry error tracking
-  Invite system (Resend + copy link)
-  Command Center ⌘K

In Progress / Near-term

-  Signal Engine cron activation (CRON_SECRET deployment)
-  Auth email delivery verification (Postmark)
-  Gate audits for Orders, Admin Console, AI Brief

Planned (Post-Beta)

-  Calendar integration
-  Business Memory Layer (persistent AI memory per workspace)
-  Team isolation V2 (RLS per-workspace toggle)
-  Public REST API
-  Outlook / Microsoft 365 email integration
-  Mobile-optimized views
-  Webhooks for external integrations

Pricing Model (Draft)

Tier	Price	What's Included
Free	€0/month	Up to 200 contacts, basic CRM, tasks, notes — no AI, no signals
Standard	€29/user/month	Unlimited CRM, Signal Engine, Relationship Health, AI (limited budget)

Tier	Price	What's Included
Pro	€79/user/month	Full AI, orders, teams, AI Brief, tag intelligence, email write, priority support
Enterprise	Custom	Admin Console, audit log, custom AI budget, team isolation, SLA, SSO

Why Gunimi

vs. HubSpot / Pipedrive

HubSpot and Pipedrive are CRMs first — they track deals and contacts but have no ambient intelligence. Gunimi observes the workspace and surfaces signals proactively. No configuration required.

vs. Notion / Obsidian

Notion is a knowledge tool. It has no concept of relationships, deals, signals, or AI that understands business context. Gunimi is built around entities and their connections.

vs. Linear / Asana

Task tools with no CRM, no relationship graph, no communication layer. Gunimi connects execution to relationships — tasks are linked to contacts, companies, deals.

The Gunimi Difference

- **One workspace, full context** — contacts, deals, tasks, notes, email, orders in one place
- **Signal Engine** — proactive intelligence, not reactive dashboards
- **Relationship Health Score** — automatic, multi-signal scoring per contact
- **AI that understands business context** — not a chatbot, but an embedded intelligence layer
- **Enterprise architecture** — RLS, audit logs, AI budget controls, admin console
- **Built for Europe** — Slovak/Czech/English, EUR currency, GDPR-aware architecture

Technical Blueprints

Gunimi maintains formal architecture blueprints in [/docs/blueprints/](#):

- [SIGNAL_ENGINE_BLUEPRINT.md](#) — 21 chapters, 25 signal types, Signal Identity contract
- [BUSINESS_MEMORY_BLUEPRINT.md](#) — 23 chapters, 7 memory types, AI Trust Boundary
- [AI_PLATFORM_ARCHITECTURE.md](#) — 15 chapters, 11 layers, 20 invariants
- [WORKSPACE_LIFECYCLE.md](#) — 4 workspace states, entry/exit conditions
- [DOMAIN_FOUNDATION_BLUEPRINT.md](#) — People graph architecture

These blueprints are living documents — they precede implementation and govern architectural decisions.

Engineering Quality Standards

Every feature must pass all quality gates before shipping:

Gate	Standard
TypeScript	<code>npm run type-check</code> — zero errors
ESLint	<code>npm run lint --max-warnings=0</code> — zero errors, zero warnings
Production Build	<code>npm run build</code> — clean compilation
Localization	Every visible string in EN + SK + CS
GDL Compliance	Uses Gunimi Design Language components
Runtime Audit	DevTools console clean, no unhandled errors

Production code standards:

- No `console.log`, `console.debug`, `console.info` anywhere
- No `console.error` in client components
- All caught errors in client code use `catch { }` without logging
- All server errors routed to Sentry via `logger.error()`

Gunimi — Build a workspace that knows your business.

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